

12.163

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Question:

The geometric transformation specified by

$$\begin{pmatrix} X' & Y' & 1 \end{pmatrix} = \begin{pmatrix} X & Y & 1 \end{pmatrix} \begin{pmatrix} 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \\ 1 & 2 & 1 \end{pmatrix}$$

in a 2D CAD system represents

- 1) Scaling and Translation
- 2) Scaling and Rotation
- 3) Rotation and Translation
- 4) Rotation

Solution:

A 2D affine transformation is of the form

$$\begin{pmatrix} \mathbf{x}'^T & 1 \end{pmatrix} = \begin{pmatrix} \mathbf{x}^T & 1 \end{pmatrix} \mathbf{T} \quad (1)$$

where the transformation matrix is

$$\mathbf{T} = \begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{t}^T & 1 \end{pmatrix}$$

(1) simplifies to

$$\begin{pmatrix} \mathbf{x}'^T & 1 \end{pmatrix} = \begin{pmatrix} \mathbf{x}^T & 1 \end{pmatrix} \begin{pmatrix} \mathbf{A} & \mathbf{0} \\ \mathbf{t}^T & 1 \end{pmatrix} \quad (2)$$

$$= \begin{pmatrix} \mathbf{x}^T \mathbf{A} + \mathbf{t}^T & 1 \end{pmatrix} \quad (3)$$

$$\mathbf{x}'^T = \mathbf{x}^T \mathbf{A} + \mathbf{t}^T \quad (4)$$

$$\mathbf{x}' = \mathbf{A}^T \mathbf{x} + \mathbf{t} \quad (5)$$

The given transformation matrix is:

$$\mathbf{T} = \begin{pmatrix} 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \\ 1 & 2 & 1 \end{pmatrix} \quad (6)$$

From this, we identify the linear transformation matrix $\mathbf{A}$ and the translation vector $\mathbf{t}$.

$$\mathbf{A} = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.25 \end{pmatrix}, \quad \mathbf{t} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad (7)$$

Since $\mathbf{A}$ is a diagonal matrix and not a multiple of the identity, it represents a non-uniform scaling. Since $\mathbf{t} \neq \mathbf{0}$, there is a translation.

A pure rotation requires the linear part $\mathbf{A}$ to be an orthogonal matrix, where $\mathbf{A}\mathbf{A}^\top = \mathbf{I}$. We check this condition:

$$\mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.25 \end{pmatrix} \begin{pmatrix} 0.5 & 0 \\ 0 & 0.25 \end{pmatrix} = \begin{pmatrix} 0.25 & 0 \\ 0 & 0.0625 \end{pmatrix} \neq \mathbf{I} \quad (8)$$

Since $\mathbf{A}$ is not orthogonal, the transformation is not a rotation.

Example: Applying the transformation to the point $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$. In homogeneous coordinates, using (5)

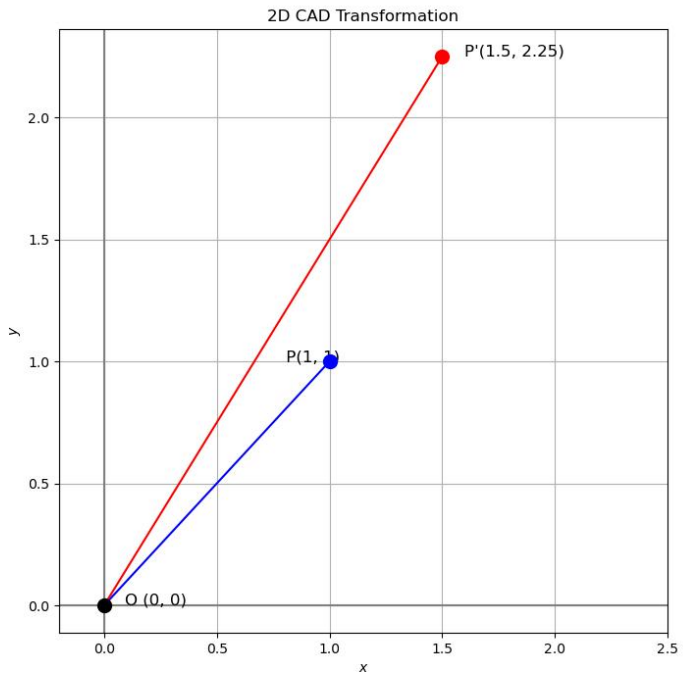
$$\mathbf{x}' = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.25 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad (9)$$

$$= \begin{pmatrix} 0.5 \\ 0.25 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad (10)$$

$$= \begin{pmatrix} 1.5 \\ 2.25 \end{pmatrix} \quad (11)$$

The point $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is transformed to $\begin{pmatrix} 1.5 \\ 2.25 \end{pmatrix}$, demonstrating both scaling and translation.

The correct option is **1) Scaling and Translation.**



Plot